EXNO :- 11

Date :- 25.8.25

Routing at Network Layer

a] Simulate Static Routing Protocol Configuration using Cisco Packet Tracer.

Aim :-

To simulate and Configure Static Routing Protocol using Cisco Packet Tracer and observe packet routing between different networks.

Procedure :-

* Open Cisco Packet Tracer.
* place three routers, two switches and four PCs.
* Connect the devices using appropriate Ethernet cabel
* Enable the router interface using the no shoutdown comma
* Configure static routes on the routers using ip route Command
* Configure a primary route and backup route using different administrative distance.
* Verify using routing table : show ip route static
* Test Connectivity between PCs using the Ping Command
* use Tracert <destination IP> to observe distination
* Simulate the failure the main route and verify that the backup route is used.

Result :-

Thus, the static Routing code successfully simulated and configured using cisco Packet Tracer through primary and backup routes was verified.

5] Simulate RIP using Packet Tracer.

Aim:

To Simulate RIP using Cisco Packet Tracer and Observee dynamic routing between different network.

Procedure:

1] Open Cisco Packet Tracer.
2] Place Three router and Two PCs.
3] Connect The Three routers in a triangular topology.
4] Connect the PC0 to Router 0 and PC1 to Router 2.
5] Configure the IPaddresses of all Router interface and PCs
6] Enable the interface using no shutdown
7] Configure RIP on each router using router rip
8] Advertise the directly connected networks using the network Command
9] Verify the routing tables using :- show ip route
10] Test Connectivity between PC0 and PC1 using
11] use Tracert to Observee the route selected by RIP
12] Disconnected one route and Observee the RIP automated selected another available route.

Result:

Thus, RIP was successfully simulated using Cisco Packet Tracer, and Dynamic route select and automatic rerouting were verified.